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ORIGINAL ARTICLE

Conventional and microwave-assisted synthesis of new indole-tethered benzimidazole-based 1,2,3-triazoles and evaluation of their antimycobacterial, antioxidant and antimicrobial activities

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Abstract

A new series of triheterocycles containing indole–benzimidazole-based 1,2,3-triazole hybrids have been synthesized in good yields via a microwave-assisted click reaction. All the compounds were characterized by IR, ^{1}H NMR, ^{13}C NMR and mass spectroscopy and were evaluated for their in vitro antitubercular activity against the *Mycobacterium tuberculosis* H37Rv strain. Compounds **4b**, **4h** and **4i** displayed highly potent antitubercular activity with MIC 3.125–6.25 µg/mL. The antioxidant potential was evaluated using 2,2-diphenyl-1-picryl hydrazine and H_2O_2 radical scavenging activity, and compounds **4e**,**4f** and **4g** showed excellent radical scavenging activity with IC_{50} values in the range of 08.50–10.05 µg/mL. Furthermore, the compounds were evaluated for antimicrobial activity against numerous bacterial and fungal strains, and compounds **4b**, **4c** and **4h** were found to be the most promising potential antimicrobial molecules with MIC 3.125–6.25 µg/mL.

Keywords Indole · Benzimidazole · 1, 2, 3-Triazole · Click chemistry · Microwave irradiation · Antitubercular · Antioxidant · Antimicrobial

Introduction

Mycobacterium tuberculosis (MTB) bacterium causes the infectious disease tuberculosis (TB) by affecting the lungs. In latent tuberculosis, the MTB presents in the body in an inactive form and does not show any symptoms and is not contagious. Active TB infection occurs much higher in persons suffering from other impaired immune systems like HIV and malnutrition [1]. Thus, as resistant strains of Mycobacterium tuberculosis (MTB) have slowly emerged, treatment failure is the main reality. However, various drugs such as streptomycin, rifampicin, isoniazid, ethionamide, ethambutol and pyrazinamide are employed for the treatment of tuberculosis. Since these drugs take more than six months to cure TB, severe side effects and attaining multidrug resistance by bacteria toward chemotherapeutics of available marketed drugs demanded the search for new chemical entities which can serve better treatment against tuberculosis [2–4].

The presence of free radicals in the body leads to oxidative stress which causes neurodegenerative diseases including Alzheimer's disease and Parkinson's disease. Other disorders include chronic fatigue syndrome, rheumatoid arthritis and blood vessel disorders such as heart attacks and atherosclerosis [5,6]. Thus, treatment with antioxidants is potentially a way to overcome oxidative stress. On the other hand, antimicrobial therapy has long been facing the problem of resistance of microorganisms. Multidrug resistance by bacterial strains claims for the search of new antimicrobial agents, which have a broad spectrum of activity against the resistant microorganisms.

Benzimidazole is an important pharmacophore in drug discovery process since it exhibits various categories of therapeutic properties such as antibacterial, antioxidants, antitubercular, anticancer, antitumor [7–9], antiamoebic, hormone modulators, anti-inflammatory [10–12] and antiviral activity [13]. Benzimidazole derivatives are known to exhibit anticancer activity and high levels of cytotoxicity against cancer cell lines [14]. In addition, benzimidazole derivatives can bind to biomolecules in living systems due to their struc-

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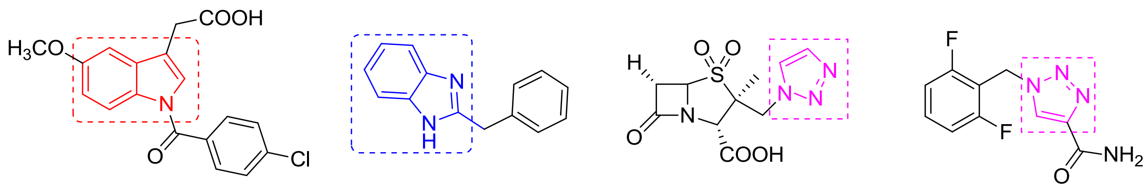


Indomethacin (Anti-inflammatory) | Bendazol (Antispasmodic) | Tazobactam (β-lactum antibiotic) | Rufinamide (Anticonvulsant)

Fig. 1 Commercially available indole, benzimidazole and 1,2,3-triazole drugs

tural similarity with some naturally occurring compounds. Indomethacin is used as anti-inflammatory drug comprised of indole nucleus, and bendazol is used as antispasmodic drug comprised of a benzimidazole nucleus (Fig. 1).

1,2,3-Triazoles are an important class of bioactive molecules due to their wide range of applications as pharmaceutical agents [15]. A large variety of therapeutic agents are comprised of 1,2,3-triazole nucleus in it, such as antifungal [16], antibacterial [17], antiallergic [18], anti-HIV [19], antitubercular [20] and anti-inflammatory agents [21]. Tazobactam is a β-lactam antibiotic drug, and rufinamide is an anticonvulsant drug containing the 1,2,3-triazole motif (Fig. 1). Microwave-assisted synthesis has gained much interest due to the rapid synthesis of heterocyclic compounds, which are not easily accessible by conventional synthetic techniques. More recently, this way we extended our research program to synthesize bioactive heterocyclic compounds [22,23]. Reports for the synthesis of benzimidazole-based 1,2,3-triazoles under microwave irradiation conditions are scare to find in the literature compared to conventional methods. Microwave irradiation offers rapid, simple and eco-friendly parameters for the synthesis of a variety of organic molecules in higher yields and shorter reaction times.

Based on the pharmacological profile of indole, benzimidazole and 1,2,3-triazole scaffolds, and the advantages of microwave technology, it is worthwhile to design and synthesize new prototypes with indole, benzimidazole and 1,2,3-triazole scaffolds in a single molecular framework. In continuation of our endeavors toward the development of anti-infective agents [24,25], the potency to exhibit antitubercular, antioxidant and antimicrobial activities of final compounds was reported.

Results and discussion

Chemistry

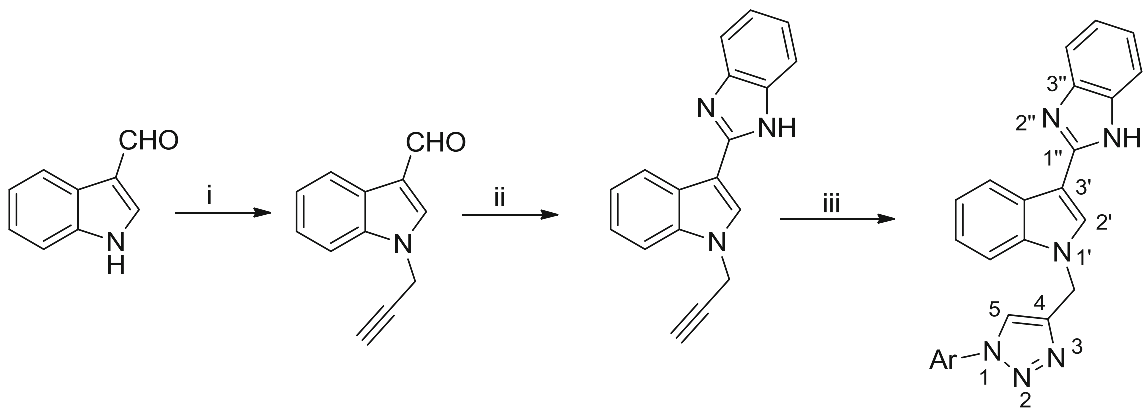
The sequence of the reactions employed for the construction of novel indole–benzimidazole-based 1,2,3-triazole hybrids is demonstrated in Scheme 1. Compound **2** was synthesized by treating propargyl bromide with 1*H*-indole-3-carbaldehyde (**1**) in the presence of potassium carbonate in dry acetone at reflux temperature [26]. Compound **2** was further reacted by heating at reflux temperature with *o*-phenylenediamine (OPDA) in chloroform for 6 h afforded compound **3**. Compound **3** coupled with variously substituted azides by azide–alkyne cycloaddition reaction to afford title compounds **4a–j**. The variously substituted azides were achieved following a reported protocol [27,28]. In order to develop a high-yield synthetic protocol for the synthesis of title compounds **4a–j**, a preliminary study was carried out for the synthesis of compound **4f** in the presence of different catalysts and solvents under conventional and microwave irradiation conditions (Table 1). The reaction was carried out in different solvent mixtures including THF:H_2O, EtOH:H_2O and DMF:H_2O in the presence of two Cu catalysts, namely (i) $CuSO_4.5H_2O$ with sodium ascorbate and (ii) CuI alone. Based on our preliminary results, the reaction using CuI in DMF:H_2O (1:3) under microwave irradiation gave better yield. In addition, a comparison study of the reaction using CuI in DMF:H_2O (1:3) under conventional and microwave irradiation condition demonstrates that the reaction time is reduced from 12 h to 8 min, respectively, and the yield is increased from 40–52% to 76–85%, respectively (Table 2).

The formation of all the title compounds was confirmed by IR, ^{1}H NMR, ^{13}C NMR and mass spectral analysis. The IR spectra of **4f** displayed a characteristic N–H absorption band at 3060 cm^{-1} and C=N stretching at 1580 cm^{-1}. The ^{1}H NMR spectrum of **4f** exhibited a broad singlet at δ 12.55 corresponding to the NH proton, a singlet at δ 8.62 was assigned to the triazole-H, a singlet at δ 5.68 corresponds to the NCH_2 protons and a singlet at δ 3.82 belongs to the OCH_3 protons. In the ^{13}C-NMR spectrum, the OCH_3 carbon appeared at δ 55.7 and the NCH_2 carbon at δ 40.6. In the mass spectrum, the base signal at m/z 421 appeared as $[M+H]^+$.

Biological activity

Antitubercular activity

The title compounds were screened in vitro for their inhibitory potency against *M. tuberculosis* H37Rv using



1 **2** **3** **4**

4a: Ar = phenyl
4b: Ar = 2-chlorophenyl
4c: Ar = 4-chlorophenyl
4d: Ar = 3-methylphenyl
4e: Ar = 4-methylphenyl
4f: Ar = 2-methoxyphenyl
4g: Ar = 4-methoxyphenyl
4h: Ar = 2-nitrophenyl
4i: Ar = 3-(trifluoromethyl)phenyl
4j: Ar = benzyl

Reagents and conditions: (i) propargyl bromide, K_2CO_3, dry acetone, reflux, 4 h;
(ii) OPDA, AcOH, $CHCl_3$, reflux, 6 h;
(iii) Ar-N_3, CuI, DMF/H_2O (1:3), 80 ^oC, 12 h or MWI, 180 W, 8 min

Scheme 1 Synthetic route for the indole–benzimidazole-based 1,2,3-triazole hybrids

Table 1 Preliminary catalyst and solvent study for the synthesis of compound **4f**

Entry	Catalyst	Solvent	Conventional heating		Microwave irradiation	
			Time (h)	Yield (%)	Time (min)	Yield (%)
4f1	$CuSO_4 \cdot 5H_2O$, sodium ascorbate	THF:H_2O (1:2)	36	16	15	18
4f2	$CuSO_4 \cdot 5H_2O$, sodium ascorbate	THF:H_2O (1:3)	36	14	15	19
4f3	$CuSO_4 \cdot 5H_2O$, sodium ascorbate	EtOH:H_2O (1:2)	24	17	15	24
4f4	$CuSO_4 \cdot 5H_2O$, sodium ascorbate	EtOH:H_2O (1:3)	24	20	15	32
4f5	$CuSO_4 \cdot 5H_2O$, sodium ascorbate	DMF:H_2O (1:2)	24	31	15	35
4f6	$CuSO_4 \cdot 5H_2O$, sodium ascorbate	DMF:H_2O (1:3)	24	34	15	40
4f7	CuI	THF:H_2O (1:2)	24	16	10	24
4f8	CuI	THF:H_2O (1:3)	24	20	10	32
4f9	CuI	EtOH:H_2O (1:2)	24	32	10	41
4f10	CuI	EtOH:H_2O (1:3)	24	36	10	47
4f11	CuI	DMF:H_2O (1:2)	12	40	8	55
4f12	CuI	DMF:H_2O (1:3)	12	42	8	77

Table 2 Comparison of yields of compounds **4a–j** under different synthetic conditions

Compound	Conventional		MWI	
	Time (h)	Yield (%)	Time (min)	Yield (%)
4a	12	51	8	81
4b	12	52	8	76
4c	12	47	8	80
4d	12	40	8	85
4e	12	51	8	79
4f	12	42	8	77
4g	12	50	8	82
4h	12	50	8	80
4i	12	45	8	76
4j	12	52	8	77

the resazurin microtiter assay (REMA). Rifampicin and isoniazid were used as positive controls. The in vitro antimycobacterial test results were expressed as minimal inhibition concentration (MIC) in μg/mL, or with respect to the molecular weight of final products in μM (values in parentheses) and are tabulated in Table 3. Amongst the series, compound **4h** emerged as best antitubercular drug candidate by inhibiting the growth of the MTB strain with MIC = 3.125 μg/mL (7.1 μM) (control rifampicin MIC = 0.04 μg/mL and isoniazid MIC = 0.38 μg/mL). The best activity of **4h** may be attributed to the presence of nitro group on phenyl ring at ortho position and was highly favored for antitubercular activity. Compound **4b** (MIC = 6.25 μg/mL (14.7 μM)) with chloro substitution, compound **4i** (MIC = 6.25 μg/mL (14.2 μM)) with trifluoromethyl substitution and compound **4j** (MIC = 12.5 μg/mL (28.4 μM)) with benzyl substitution exhibited moderate antitubercular activity. Notably, the incorporation of the electron-withdrawing nitro group and electronegative Cl and trifluoromethyl groups on the phenyl ring was highly favored for antitubercular activity.

We calculated various physicochemical parameters such as clogP, drug score and drug-likeness of **4a–j** using the Osiris Property Explorer software [29]. For all the compounds, the calculated clogP values were found to be below 5 according to Lipinski's rule-of-5 [30]. The majority of the commercial drugs show positive drug score values. As can be observed in Table 3, our theoretical data showed that compounds **4a–j** also exhibited positive values for drug score.

Antioxidant activity

The antioxidant ability of the new library was evaluated by DPPH and H_2O_2 radical scavenging assay. Natural antioxidant ascorbic acid and synthetic antioxidant BHT (butylated hydroxytoluene) were employed as standard antioxidants. Most of the compounds potentially inhibited DPPH radical compared to the standard (Table 4). Compounds **4e**, **4f** and **4g** with 4-methyl, 2-methoxy and 4-methoxy substitution on phenyl ring, respectively, showed potent radical scavenging ability on the DPPH, with IC_{50} values of 10.04 ± 1.10, 8.52 ± 0.57 and 9.25 ± 0.85 μg/mL, respectively. These results were lower than the standards (AA and BHT with 13.62 ± 0.69 and 17.16 ± 0.25 μg/mL, respectively), indicating that compounds **4e**, **4f**, and **4g** emerged as highly potent antioxidant molecules toward DPPH. These results demonstrate that the presence of methyl and methoxy substituents at ortho and para positions of phenyl nucleus played a significant role in exhibiting the antioxidant activity.

The same series was screened for H_2O_2 radical scavenging potency. In Table 4, compounds **4e**, **4f** and **4g** exhibited highest H_2O_2 scavenging effect with potent IC_{50} values of 11.19 ± 0.60, 10.18 ± 0.75 and 10.76 ± 0.42 μg/mL, respec-

Table 3 Antimycobacterial activity (MIC in μg/mL, and μM) and pharmacological parameters for bioavailability of compounds **(4a–j)**

Compound	*M. tuberculosis* H37Rv*	clogP#	Drug-likeness#	Drug score#
4a	25 (32.0)	3.30	0.10	0.50
4b	6.25 (14.7)	3.91	– 0.14	0.38
4c	12.5 (29.4)	3.91	0.31	0.41
4d	25 (61.8)	3.64	– 0.51	0.40
4e	>50	3.64	– 1.73	0.34
4f	>50	3.23	– 0.39	0.44
4g	50(119.0)	3.23	– 1.76	0.36
4h	3.125 (7.1)	2.38	– 5.47	0.30
4i	6.25 (14.2)	4.15	– 6.76	0.24
4j	12.5 (28.4)	3.55	– 0.16	0.47
Rifampicin	0.04 (0.04)	2.26	10.16	0.20
Isoniazid	0.38 (2.7)	– 1.02	– 5.06	0.06

*Micromolar values in parentheses Conversion formula: MIC(μM) = [{MIC(μg/mL)÷MW}×1000]
#Obtained from molecular Osiris Property Explorer

Table 4 Antioxidant activity of compounds (**4a–j**)

Compound	Scavenging activity ($IC_{50}\mu g/mL$)	
	DPPH	H_2O_2
4a	14.45 ± 1.04	15.61 ± 0.54
4b	15.32 ± 0.45	17.05 ± 0.71
4c	18.54 ± 0.58	18.24 ± 0.49
4d	14.08 ± 0.42	16.52 ± 1.20
4e	10.04 ± 1.10	11.19 ± 0.60
4f	08.52 ± 0.57	10.18 ± 0.75
4g	09.25 ± 0.85	10.76 ± 0.42
4h	22.72 ± 0.76	28.12 ± 0.84
4i	24.40 ± 0.29	23.25 ± 0.93
4j	21.07 ± 0.40	18.53 ± 0.75
AA*	13.62 ± 0.69	16.83 ± 0.48
BHT#	17.16 ± 0.25	17.62 ± 0.87

*Ascorbic acid

#Butylated hydroxytoluene

Lower IC_{50} values indicate better radical scavenging ability

tively. The rest of the compounds exhibited least-to-moderate antioxidant activity. It is obvious from the screening of antioxidant activity that the compounds with methyl and methoxy groups on the phenyl ring were highly favored for radical scavenging activity.

Antimicrobial activity

The series **4a–j** was tested for their antibacterial activity against gram-positive *Staphylococcus aureus* (ATCC 6538), *Bacillus subtilis* (ATCC 6633) and gram-negative *Proteus vulgaris* (ATCC 29213), *Escherichia coli* (ATCC 11229). Gentamicin was used as standard, and the activity results were obtained as minimum inhibitory concentration (MIC) in µg/mL (Table 5). As observed from Table 5, compounds **4b** (3.125–6.25 µg/mL) (Ar = 2-chlorophenyl), **4c** (3.125–6.25 µg/mL) (Ar = 4-chlorophenyl) and **4h** (3.125–6.25 µg/mL) (Ar = 2-nitrophenyl) those have electron-withdrawing nitro group and electronegative substituents on the phenyl ring exhibited more potent inhibition against bacterial strains. On the other hand, methyl and methoxy substituents exhibited poor antimicrobial activity. It can be concluded that the nitro group and chloro substituents on the phenyl ring have strong effect on antibacterial activity [31].

The same library was tested for the antifungal activity against *Candida albicans* (ATCC 10231) and *Aspergillus niger* (ATCC 9029). In Table 5, compounds **4b** (Ar = 2-chlorophenyl), **4c** (Ar = 4-chlorophenyl) and **4h** (Ar = 2-nitrophenyl) were exhibited moderate antifungal activity with the MIC 6.25, 6.25 – 12.5 and 6.25 µg/mL, respectively. The remaining compounds exhibited poor antifungal ability.

Conclusion

In summary, we synthesized a novel library of indole–benzimidazole-based 1,2,3-triazole derivatives under conventional and microwave irradiation conditions. The microwave irradiation method was proved to be highly efficient as the reactions completed with high yields in short reaction time. All the target compounds were tested for antimycobacterial activity, and compounds **4b**, **4h** and **4i** exhibited significant potency. The antioxidant activity screening revealed that compounds **4e**, **4f** and **4g** were found to be potent radical scavenging agents. In addition, the antimicrobial activity of the title compounds demonstrated that compounds **4b**, **4c** and **4h** were more potent against pathogenic bacteria and fungi.

Table 5 Antimicrobial activity (MIC in µg/mL) of newly synthesized compounds

Compound	Gram-positive bacteria		Gram-negative bacteria		Fungal strains	
	B. subtilis	*S. aureus*	*E. coli*	*P. vulgaris*	*A. niger*	*C. albicans*
4a	25	25	50	25	12.5	25
4b	3.125	6.25	6.25	6.25	6.25	6.25
4c	6.25	3.125	6.25	6.25	12.5	6.25
4d	50	25	25	12.5	50	12.5
4e	25	50	25	25	50	50
4f	50	50	50	25	100	50
4g	12.5	12.5	25	50	50	100
4h	3.125	3.125	3.125	6.25	6.25	6.25
4i	50	25	50	50	25	25
4j	12.5	12.5	25	12.5	25	50
Gentamicin	1.56	1.56	1.56	3.125	–	–
Fluconazole	–	–	–	–	3.125	1.56
DMSO	–	–	–	–	–	–

Experimental

Chemistry

All chemicals and reagents used in this study were obtained from Sigma-Aldrich and were used as received without further purification unless mentioned. A Shimadzu FTIR 8400 S spectrometer was employed to record the IR spectra $\tilde{\upsilon}$ in cm^{-1} (KBr). Melting points were recorded on Stuart SMP3 melting-point apparatus and are uncorrected. Mass spectra (EI) were recorded on a Finnigan MAT 1020 mass spectrometer, in m/z. A Bruker Avance-400 spectrometer was employed to record the ^{1}H NMR and ^{13}C NMR spectra using tetramethylsilane as standard and $CDCl_3$, DMSO-d6 as solvents. In ^{1}H NMR spectra, the signals were recorded as singlet (s), doublet (d), triplet (t), quartet (q) and multiplet (m). Karlo Erba 1106 elemental analyzer was used to record the elemental analyses. Silica gel percolated TLC plates of Merck 60 F254 were employed to monitor the reaction progress, and spots were visualized with UV light. CEM Discover microwave system was used to carry out microwave irradiation experiments. The reaction temperature in microwave irradiation experiments was monitored using an IR sensor.

General procedure for the synthesis of 1-(prop-2-yn-1-yl)-1*H*-indole-3-carbaldehyde (2)

To a stirred solution of 1*H*-indole-3-carbaldehyde (**1**) (145 mg, 1 mmol) in dry acetone (5mL) in the presence of anhydrous K_2CO_3 (210 mg, 1.5 mmol), propargyl bromide (118 mg, 1 mmol) was added and the reaction mixture was refluxed for 4 h. After completion of the reaction, the solvent was evaporated under vacuum, and the solid residue was purified by column chromatography using hexane/ethyl acetate (7:3) as eluent to afford 1-(prop-2-yn-1-yl)-1*H*-indole-3-carbaldehyde (**2**). Yield: 76 %; M.P.: 128 °C.

General procedure for the synthesis of 2-(1-(prop-2-yn-1-yl)-1*H*-indol-3-yl)-1*H*-benzo[*d*]imidazole (3)

To a well-stirred solution of 1-(Prop-2-yn-1-yl)-1*H*-indole-3-carbaldehyde (**2**) (183 mg, 1 mmol) in $CHCl_3$ (5 mL), few drops of AcOH and *o*-Phenylenediamine (108 mg, 1 mmol) were added and refluxed for 6 h. After completion of reaction (monitored by TLC), the reaction mixture was poured into ice-cold water, then extracted with ethyl acetate and dried over Na_2SO_4 and column purified using hexane/ethyl acetate (4:1) to afford 2-(1-(prop-2-yn-1-yl)-1*H*-indol-3-yl)-1*H*-benzo[*d*]imidazole (**3**). Yield: 84 %; M.P.: 142 °C.

General procedure for synthesis of compounds (4a–j)

Conventional method

To a mixture of compound **3** (1 mmol) and azide (1 mmol) in DMF/H_2O (1:3) (20 mL), CuI (9.5 mg, 0.05 mmol) was added and heated for 12 h at 80 °C. After completion of reaction (as indicated by TLC), the reaction mixture was added to 30 mL of ice-cold water, washed twice with brine. The crude was extracted with ethyl acetate, concentrated under reduced pressure and column purified using hexane/ethyl acetate (3:1) to afford compound **4**.

Microwave-assisted synthesis

To a mixture of compound **3** (1 mmol) and azide (1 mmol) in DMF/H_2O (1:3) (4 mL), CuI (9.5 mg, 0.05 mmol) was added. The resulting mixture was capped in a closed vessel and subjected to microwave irradiation at 180 W for 8 min. After completion of reaction (as indicated by TLC), the reaction mixture was added to 30 mL of ice-cold water, washed twice with brine. The crude was extracted with ethyl acetate, concentrated under reduced pressure and column purified using hexane/ethyl acetate (3:1) to afford compound **4**.

2-(1-((1-Phenyl-1*H*-1,2,3-triazol-4-yl)methyl)-1*H*-indol-3-yl)-1*H* -benzo[*d*]imidazole (4a)

Pale white solid; m.p.: 120–122 °C. IR (KBr, cm^{-1}): 3125 (N–H), 1576 (C=N). ^{1}H-NMR (400 MHz, DMSO-d_6) δ: 12.48 (bs, 1H, NH), 8.60 (s, 1H, triazole-H), 8.57 (d, 1H, $J = 7.46$ Hz, Ar–H), 8.32 (s, 1H, indole-H), 7.79–7.77 (m, 2H, Ar–H), 7.59 (d, 1H, $J = 7.52$ Hz, Ar–H), 7.56–7.54 (m, 3H, Ar–H), 7.34–7.28 (m, 3H, Ar–H), 7.15–7.08 (m, 3H, Ar–H), 5.65 (s, 2H, NCH_2). ^{13}C-NMR (100 MHz, DMSO-d_6) δ: 151.42 (C-1″), 146.69 (C-3″), 143.44 (C–Ar), 136.40 (C-9′), 131.21 (C-4), 128.49 (C-2′), 126.84 (C–Ar), 125.41 (C–Ar), 122.54 (C–Ar), 121.32 (C–Ar), 120.67 (C–Ar), 120.35 (C–Ar), 112.20 (C-5), 110.19 (C–Ar), 105.47 (C-3′), 41.72 (C, NCH_2). LC-MS: m/z 390 $(M)^{+}$. Anal. Calc. for $C_{24}H_{18}N_6$: C, 73.83; H, 4.65; N, 21.52. Found: C, 73.86; H, 4.67; N, 21.56.

2-(1-((1-(2-Chlorophenyl)-1*H*-1,2,3-triazol-4-yl)methyl)-1*H*-indol-3-yl)-1*H*-benzo[*d*]imidazole (4b)

Pale white solid; m.p.: 123–125 °C. IR (KBr, cm^{-1}): 3112 (N–H), 1569 (C=N). ^{1}H-NMR (400 MHz, $CDCl_3$) δ: 10.01 (bs, 1H, NH), 8.25–8.23 (m, 1H, Ar–H), 7.89 (s, 1H, triazole-H), 7.63 (s, 1H, indole-H), 7.52–7.48 (m, 3H, Ar–H), 7.45–7.32 (m, 4H, Ar–H), 7.23–7.20 (m, 2H, Ar–H), 7.15–7.12 (m, 2H, Ar–H), 5.29 (s, 2H, NCH_2). ^{13}C-NMR

(100 MHz, $CDCl_3$) δ: 148.39 (C-1″), 143.09 (C-3″), 137.67 (C–Ar), 136.31 (C-9′), 134.43 (C-4), 130.97 (C-2′), 130.67 (C–Ar), 129.53 (C–Ar), 128.66 (C–Ar), 127.88 (C–Ar), 127.71 (C–Ar), 125.50 (C–Ar), 124.45 (C–Ar), 123.09 (C–Ar), 122.44 (C–Ar), 121.60 (C–Ar), 120.78 (C–Ar), 114.21 (C-5), 110.17 (C–Ar), 105.88 (C-3′), 41.85 (C, NCH_2). LC-MS: m/z 425 $(M+H)^+$. Anal. Calc. for $C_{24}H_{17}ClN_6$: C, 67.84; H, 4.03; N, 19.78. Found: C, 67.87; H, 4.06; N, 19.82.

2-(1-((1-(4-Chlorophenyl)-1*H*-1,2,3-triazol-4-yl)methyl)-1*H*-indol-3-yl)-1*H*-benzo[*d*]imidazole (4c)

White solid; m.p.: 140–143 °C. IR (KBr, cm^{-1}): 3118 (N–H), 1552 (C=N). ^{1}H-NMR (400 MHz, $CDCl_3$) δ: 8.27–8.25 (d, 1H, J = 8.03 Hz, Ar–H), 7.86 (s, 1H, triazole-H), 7.61 (s, 1H, indole-H), 7.56–7.53 (m, 2H, Ar–H), 7.49–7.47 (d, 2H, J = 8.78 Hz, Ar–H), 7.43–7.41 (d, 2H, J = 7.02 Hz, Ar–H), 7.38–7.36 (d, 3H, J = 8.78 Hz, Ar–H), 7.20–7.18 (m, 2H, Ar–H), 5.31 (s, 2H, NCH_2). ^{13}C-NMR (100 MHz, DMSO-d_6) δ: 151.43 (C-1″, 145.89 (C-3″), 144.29 (C–Ar), 136.41 (C-9′), 132.32 (C-4), 128.25 (C-2′), 126.27 (C–Ar), 124.75 (C–Ar), 123.61 (C–Ar), 121.90 (C–Ar), 121.07 (C–Ar), 121.13 (C–Ar), 112.30 (C-5), 111.18 (C–Ar), 105.54 (C-3′), 41.71 (C, NCH_2). LC-MS: m/z 425 $(M+H)^+$. Anal. Calc. for $C_{24}H_{17}ClN_6$: C, 67.84; H, 4.03; N, 19.78. Found: C, 67.88; H, 4.06; N, 19.81.

2-(1-((1-(*m*-Tolyl)-1*H*-1,2,3-triazol-4-yl)methyl)-1*H*-indol-3-yl)-1*H*-benzo[*d*]imidazole (4d)

Pale white solid; m.p.: 135–137 °C. IR (KBr, cm^{-1}): 3122 (N–H), 1563 (C=N); ^{1}H-NMR (400 MHz, $CDCl_3$) δ: 8.23–8.21 (m, 1H, Ar–H), 7.91 (s, 1H, triazole-H), 7.68 (s, 1H, indole-H), 7.44–7.42 (m, 3H, Ar–H), 7.39 (s, 1H, Ar–H), 7.30–7.28 (d, 3H, J = 9.78 Hz, Ar–H), 7.17–7.15 (t, 4H, J = 3.26 Hz, J = 9.03 Hz, Ar–H), 5.12 (s, 2H, NCH_2), 2.33 (s, 3H, CH_3). ^{13}C-NMR (100 MHz, DMSO-d_6) δ: 148.17 (C-1″), 143.89 (C-3″), 139.95 (C–Ar), 137.17 (C-9'), 136.48 (C-4), 136.23 (C-2′), 129.92 (C–Ar), 129.67 (C–Ar), 129.45 (C–Ar), 125.33 (C–Ar), 123.12 (C–Ar), 122.58 (C–Ar), 121.64 (C-5), 121.00 (C–Ar), 120.74 (C-3′), 120.60 (C–Ar), 117.45 (C–Ar), 114.10 (C–Ar), 110.15 (C–Ar), 105.31 (C–Ar), 41.86 (C, NCH_2), 21.30 (C, $ArCH_3$). LC-MS: m/z 405 $(M+H)^+$. Anal. Calc. for $C_{25}H_{20}N_6$: C, 74.24; H, 4.98; N, 20.78. Found: C, 74.27; H, 5.02; N, 20.82.

2-(1-((1-(*p*-Tolyl)-1*H*-1,2,3-triazol-4-yl)methyl)-1*H*-indol-3-yl)-1*H*-benzo[*d*]imidazole (4e)

White solid. m.p.: 121–124 °C. IR (KBr, cm^{-1}): 3118 (N–H), 1554 (C=N). ^{1}H-NMR (400 MHz, $CDCl_3$) δ: 8.27–8.25 (d, 1H, J = 8.03 Hz, Ar–H), 7.86 (s, 1H, triazole-H), 7.59 (s, 1H, indole-H), 7.53–7.50 (m, 2H, Ar–H), 7.40–7.38 (d, 1H, J = 8.28 Hz, Ar–H), 7.36–7.34 (d, 1H, J = 6.52 Hz, Ar–H), 7.20–7.12 (m, 6H, Ar–H), 5.21 (s, 2H, NCH_2), 2.34 (s, 3H, CH_3). ^{13}C-NMR (100 MHz, DMSO-d_6) δ: 148.76 (C-1″), 143.97 (C-3″), 139.05 (C–Ar), 138.30 (C-9′), 136.32 (C-4), 134.24 (C-2′), 130.15 (C–Ar), 129.15 (C–Ar), 125.65 (C–Ar), 122.29 (C–Ar), 121.53 (C–Ar), 120.98 (C–Ar), 120.38 (C-5), 120.25 (C–Ar), 114.38 (C-3′), 110.10 (C–Ar), 106.44 (C–Ar), 41.89 (C, NCH_2), 21.05 (C, $ArCH_3$). LC-MS: m/z 405 $(M+H)^+$. Anal. Calc. for $C_{25}H_{20}N_6$: C, 74.24; H, 4.98; N, 20.78. Found: C, 74.28; H, 5.01; N, 20.81.

2-(1-((1-(2-Methoxyphenyl)-1*H*-1,2,3-triazol-4-yl)methyl)-1*H*-indol-3-yl)-1*H*-benzo[*d*]imidazole (4f)

White solid. m.p.: 126–128 °C. IR (KBr, cm^{-1}): 3060 (N–H), 1580 (C=N). ^{1}H-NMR (400 MHz, DMSO-d_6) δ: 12.55 (bs, 1H, NH), 8.62 (s, 1H, triazole-H), 8.56–8.54 (d, 1H, J = 7.52 Hz, Ar–H), 8.29 (s, 1H, indole-H), 7.81–7.79 (d, 1H, J = 8.03 Hz, Ar–H), 7.62–7.60 (d, 1H, J = 7.52 Hz, Ar–H), 7.55–7.49 (m, 3H, Ar–H), 7.30–7.23 (m, 3H, Ar–H), 7.18–7.10 (m, 3H, Ar–H), 5.68 (s, 2H, NCH_2), 3.82 (s, 3H, OCH_3). ^{13}C-NMR (100 MHz, DMSO-d_6) δ: 151.15 (C-1″), 148.60 (C-3″), 142.20 (C–Ar), 135.96 (C-9′), 130.43 (C-4), 128.61 (C-2′), 125.34 (C–Ar), 125.31 (C–Ar), 122.11 (C–Ar), 121.36 (C–Ar), 120.91 (C–Ar), 120.54 (C–Ar), 120.39 (C–Ar), 112.66 (C-5), 110.37 (C–Ar), 106.11 (C-3′), 55.77 (C, $O\underline{C}H_3$), 40.60 (C, $N\underline{C}H_2$). LC-MS: m/z 421 $(M+H)^+$. Anal. Calc. for $C_{25}H_{20}N_6O$: C, 71.41; H, 4.79; N, 19.99. Found: C, 71.45; H, 4.82; N, 20.04.

2-(1-((1-(4-Methoxyphenyl)-1*H*-1,2,3-triazol-4-yl)methyl)-1*H*-indol-3-yl)-1*H*-benzo[*d*]imidazole (4g)

White solid. m.p.: 129–133 °C. IR (KBr, cm^{-1}): 3150 (N–H), 1562 (C=N). ^{1}H-NMR (400 MHz, DMSO-d_6) δ: 12.52 (bs, 1H, NH), 8.82 (s, 1H, triazole-H), 8.56–8.53 (d, 1H, J = 7.02 Hz, Ar–H), 8.24 (s, 1H, indole-H), 7.80–7.72 (m, 2H, Ar–H), 7.62–7.44 (m, 3H, Ar–H), 7.30–7.25 (m, 2H, Ar–H), 7.17–7.07 (m, 4H, Ar–H), 5.66 (s, 2H, NCH_2), 3.81 (s, 3H, OCH_3). ^{13}C-NMR (100 MHz, DMSO-d_6) δ: 159.28 (C-1″), 148.89 (C-3″), 143.58 (C–Ar), 136.47 (C-9′), 129.86 (C-4), 128.84 (C-2′), 128.79 (C–Ar), 125.64 (C–Ar), 122.47 (C–Ar), 121.98 (C–Ar), 121.77 (C–Ar), 121.72 (C–Ar), 120.71 (C–Ar), 114.83 (C-5), 110.58 (C–Ar), 106.46 (C-3′), 55.52 (C, OCH_3), 41.00 (C, NCH_2). LC-MS: m/z 421 $(M+H)^+$. Anal. Calc. for $C_{25}H_{20}N_6O$: C, 71.41; H, 4.79; N, 19.99. Found: C, 71.44; H, 4.83; N, 20.03.

2-(1-((1-(2-Nitrophenyl)-1*H*-1,2,3-triazol-4-yl)methyl)-1*H*-indol-3-yl)-1*H*-benzo[*d*]imidazole (4h)

Pale white solid. m.p.: 122–125 °C. IR (KBr, cm^{-1}): 3124 (N–H), 1562 (C=N). ^{1}H-NMR (400 MHz, DMSO-d_6) δ: 8.15–8.13 (d, 1H, J = 8.03 Hz, Ar–H), 8.03–8.01 (d, 1H, J = 7.52 Hz, Ar–H), 7.90 (s, 1H, triazole-H), 7.75–7.64 (m, 3H, Ar–H), 7.54–7.48 (m, 4H, Ar–H), 7.20–7.14 (m, 4H, Ar–H), 7.18–7.14, 5.33 (s, 2H, NCH_2). ^{13}C-NMR (100 MHz, DMSO-d_6) δ: 151.81 (C-1″), 147.29 (C-3″), 144.43 (C–Ar), 136.34 (C-9′), 132.01 (C-4), 128.44 (C–Ar), 126.74 (C-2′), 126.27 (C–Ar), 123.74 (C–Ar), 121.4 (C–Ar), 121.05 (C–Ar), 120.74 (C–Ar), 112.26 (C-5), 110.64 (C–Ar), 105.88 (C-3′), 41.95 (C, NCH_2). LC-MS: m/z 436 $(M+H)^+$. Anal. Calc. for $C_{24}H_{17}N_7O_2$: C, 66.20; H, 3.94; N, 22.52. Found: C, 66.23; H, 3.97; N, 22.56.

2-(1-((1-(3-(Trifluoromethyl)phenyl)-1*H*-1,2,3-triazol-4-yl)methyl)-1*H*-indol-3-yl)-1*H*-benzo[*d*]imidazole (4i)

Pale white solid. m.p.: 128–130 °C. IR (KBr, cm^{-1}): 3112 (N–H), 1552 (C=N). ^{1}H-NMR (400 MHz, DMSO-d_6) δ: 12.28 (bs, 1H, NH), 8.52 (s, 1H, triazole-H), 8.42–8.40 (d, 1H, J = 7.62 Hz, Ar–H), 8.31 (s, 1H, indole-H), 7.81–7.77 (m, 2H, Ar–H), 7.58 (s, 1H, Ar–H), 7.56–7.54 (m, 3H, Ar–H), 7.30–7.26 (m, 2H, Ar–H), 7.19–7.15 (m, 3H, Ar–H), 5.36 (s, 2H, NCH_2). ^{13}C-NMR (100 MHz, DMSO-d_6) δ: 152.24 (C-1″), 146.48 (C-3″), 144.72 (C–Ar), 136.09 (C-9′), 131.17 (C-4), 127.81 (C–Ar), 126.44 (C-2′), 125.62 (C–Ar), 123.20 (C–Ar), 121.49 (C, CF_3), 121.08 (C–Ar), 120.77 (C–Ar), 112.45 (C-5), 111.24 (C–Ar), 105.76 (C-3′), 41.59 (C, NCH_2). LC-MS: m/z 459 $(M+H)^+$. Anal. Calc. for $C_{25}H_{17}F_3N_6$: C, 65.50; H, 3.74; N, 18.33. Found: C, 65.53; H, 3.77; N, 18.37.

2-(1-((1-Benzyl-1*H*-1,2,3-triazol-4-yl)methyl)-1*H*-indol-3-yl)-1*H*-benzo[*d*]imidazole (4j)

Pale white solid. m.p.: 131–134 °C. IR (KBr, cm^{-1}): 3118 (N–H), 1564 (C=N). ^{1}H-NMR (400 MHz, DMSO-d_6) δ: 12.35 (bs, 1H, NH), 8.62 (s, 1H, triazole-H), 8.54–8.52 (d, 1H, J = 7.56 Hz, Ar–H), 8.30 (s, 1H, indole-H), 7.76–7.74 (m, 2H, Ar–H), 7.62–7.60 (d, 1H, J = 7.42 Hz, Ar–H), 7.52–7.48 (m, 3H, Ar–H), 7.36–7.31 (m, 3H, Ar–H), 7.12–7.07 (m, 3H, Ar–H), 5.31 (s, 2H, NCH_2), 5.26 (s, 2H, Ph − CH_2). ^{13}C-NMR (100 MHz, DMSO-d_6) δ: 152.60 (C-1″), 145.39 (C-3″), 143.34 (C–Ar), 135.77 (C-9’), 131.56 (C-4), 127.20 (C–Ar), 126.44 (C-2′), 125.14 (C–Ar), 122.69 (C–Ar), 121.57 (C–Ar), 120.43 (C–Ar), 120.01 (C–Ar), 112.37 (C-5), 110.56 (C–Ar), 105.77 (C-3′), 52.71 (C, $PhCH_2$), 41.35 (C, NCH_2). LC-MS: m/z 405 $(M+H)^+$. Anal. Calc. for $C_{25}H_{20}N_6$: C, 74.24; H, 4.98; N, 20.78. Found: C, 74.27; H, 5.01; N, 20.82.

Biological assays

Antitubercular activity

The drug susceptibility and evaluation of MIC of the title compounds against *M. tuberculosis* H37Rv were performed using the resazurin microtiter assay (REMA) protocol [32]. The synthesized compounds were screened against *M. tuberculosis* H37Rv using serial dilution technique in Middlebrook 7H9 broth medium. Rifampicin and isoniazid were used as standard drugs. Each compound (2 mg) was dissolved in 1 mL of DMSO. The serial dilution of each compound was prepared using 96-well microtiter plate, and 100 mL of MTB cell suspension in nutrient media was added to each well. After 7 days of incubation, resazurin dye solution (0.02% w/v dissolved in distilled water) was added to each well and again incubated for 1 day. The MIC was determined by the minimum concentration of compound that inhibits the growth of MTB that is indicated by a color change from nonfluorescent blue to fluorescent pink color. The MIC values were calculated by visual inspection for each well showing more than 90% inhibition.

Antioxidant activity

DPPH radical scavenging activity

The radical scavenging potency of the compounds was tested from the bleaching of the purple colored methanol solution of DPPH [33]. To 4 mL of methanol solution of DPPH (0.004% (w/v)), 1 mL of 5, 10, 25, 50 and 100 µg/mL concentrations of the test compounds in methanol was added. This mixture was incubated for 30 min at room temperature, and later the absorbance was recorded against blank at 517 nm. The result was measured as the percent of inhibition of free radical production from DPPH and was calculated by the following equation.

$$\%\text{of scavenging} = [(\text{A control} - \text{A sample})/\text{A control}] \times 100$$

where A control is the absorbance of the control reaction and A sample is the absorbance of the test compound. Tests were performed in triplicate.

Hydrogen peroxide (H_2O_2) scavenging activity

Various concentrations of the test compounds (5, 10, 25, 50 and 100 µg/mL) were prepared in phosphate buffer. A solution of H_2O_2 (40 mM) was prepared in phosphate buffer (pH 7.4). The test compounds in 3.4 mL phosphate buffer were added to the H_2O_2 solution (0.6 ml, 40 mM) [34]. The absorbance value of the resulting mixture was recorded at 230 nm. The percent of inhibition of free radical production from H_2O_2 was calculated using the above equation.

Antimicrobial assay

The *in vitro* antimicrobial screening was performed by agar well diffusion method against various pathogens [35,36]. Various concentrations of test samples were prepared in DMSO. Nutrient broth (NB) plates were swabbed with 24-h-old broth culture (100 mL) of test pathogen. Using the sterile cork borer, 6-mm-depth wells were made into each petriplate. With the help of sterile pipettes, different concentrations of test compounds dissolved in DMSO were added to the wells. Along with test samples, the standard antibiotics, gentamicin, and fluconazole and DMSO as negative control were also tested against the microorganisms. The plates were incubated at 37 °C for 24 h for bacteria and at 28 °C for 48 h for fungi. After incubation, the diameter of the zone of inhibition of each well was measured. To determine MIC of the test samples, broth dilution method was employed [37,38]. The 24-h-old culture of the pathogens was diluted 100-fold in nutrient broth (100-mL test cultures in 10 mL NB). The test samples were added to the pathogen cultures in an increasing manner of their concentration. All the tubes were incubated at 37 °C for 24 h for bacteria and at 28 °C for 48 h for fungi. The tubes were examined for visible turbidity and using NB as a control. The lowest concentration that inhibited visible growth of the tested organisms was recorded as MIC.

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Compliance with ethical standards

Conflict of interest The authors confirm that this article content has no conflict of interest.

References

1. World Health Organization (WHO), Global Tuberculosis Report 2016; Available at http://www.who.int/mediacentre/factsheets/fs104/en/; accessed on 12th Mar 2017
2. Bodiang CK (2000) Issues facing TB control (2.1). Tuberculosis control in refugee populations: a focus on developing countries. Scott Med J 45:25–28
3. Long R (2000) Drug-resistant tuberculosis. Can Med Assoc J 163:425–428
4. Pozniak A (2000) HIV-associated tuberculosis in the era of HAART. Int J Tuberc Lung Dis 4:993–994
5. Cheng JH, Huang AM, Hour TC, Yang SC, Pu YS, Lin CN (2011) Antioxidant xanthone derivatives induce cell cycle arrest and apoptosis and enhance cell death induced by cisplatin in NTUB1 cells associated with ROS. Eur J Med Chem 46:1222–1231. https://doi.org/10.1016/j.ejmech.2011.01.043
6. Aldawsari FS, Aguiar RP, Wiirzler LA, Aguayo-Ortiz R, Aljuhani N, Cuman RK, Medina-Franco JL, Siraki AG, Velázquez-Martínez CA (2016) Antiinflammatory and antioxidant properties of a novel resveratrol-salicylate hybrid analog. Bioorg Med Chem Lett 26:1411–1415. https://doi.org/10.1016/j.bmcl.2016.01.069
7. Desai NC, Shihory NR, Kotadiya GM, Desai P (2014) Synthesis, antibacterial and antitubercular activities of benzimidazole bearing substituted 2-pyridone motifs. Eur J Med Chem 82:480–489. https://doi.org/10.1016/j.ejmech.2014.06.004
8. Hong SY, Kwak KW, Ryu CK, Kang SJ, Chung KH (2008) Antiproliferative effects of 6-anilino-5-chloro-1H-benzo[d]imidazole-4,7-dione in vascular smooth muscle cells. Bioorg Med Chem 16:644–649. https://doi.org/10.1016/j.bmc.2007.10.069
9. Bandyopadhyay P, Sathe M, Ponmariappan S, Sharma A, Sharma P, Srivastava AK, Kaushik MP (2011) Exploration of in vitro time point quantitative evaluation of newly synthesized benzimidazole and benzothiazole derivatives as potential antibacterial agents. Bioorg Med Chem Lett 21:7306–7309. https://doi.org/10.1016/j.bmcl.2011.10.034
10. Xu JY, Zeng Y, Ran Q, Wei Z, Bi Y, He QH, Wang QJ, Hu S, Zhang J, Tang MY, Hua WY, Wu XM (2007) Synthesis and biological activity of 2-alkylbenzimidazoles bearing a N-phenylpyrrole moiety as novel angiotensin II AT1 receptor antagonists. Bioorg Med Chem Lett 17:2921–2926. https://doi.org/10.1016/j.bmcl.2007.02.042
11. Taniguchi K, Shigenaga S, Ogahara T, Fujitsu T, Matsuo M (1993) Synthesis and antiinflammatory and analgesic properties of 2-amino-1H-benzimidazole and 1,2-dihydro-2-iminocycloheptimidazole derivatives. Chem Pharm Bull 41:301–309. https://doi.org/10.1248/cpb.41.301
12. Sondhi SM, Rajvanshi S, Johar M, Bharti N, Azam A, Singh AK (2002) Anti-inflammatory, analgesic and antiamoebic activity evaluation of pyrimido[1,6-a]benzimidazole derivatives synthesized by the reaction of ketoisothiocyanates with mono and diamines. Eur J Med Chem 37:835–43. https://doi.org/10.1016/S0223-5234(02)01403-4
13. Starcević K, Kralj M, Ester K, Sabol I, Grce M, Pavelić K, Karminski-Zamola G (2007) Synthesis, antiviral and antitumor activity of 2-substituted-5-amidino-benzimidazoles. Bioorg Med Chem 15:4419–4426. https://doi.org/10.1016/j.bmc.2007.04.032
14. Snow RJ, Abeywardane A, Campbell S, Lord J, Kashem MA, Khine HH, King J, Kowalski JA, Pullen SS, Roma T, Roth GP, Sarko CR, Wilson NS, Winters MS, Wolak JP, Cywin CL (2007) Hit-to-lead studies on benzimidazole inhibitors of ITK: Discovery of a novel class of kinase inhibitors. Bioorg Med Chem Lett 17:3660–3665. https://doi.org/10.1016/j.bmcl.2007.04.045
15. Kacprzak K (2005) Efficient one-pot synthesis of 1,2,3-triazoles from benzyl and alkyl halides. Synlett 36:943–946. https://doi.org/10.1055/s-2005-864809
16. Aher NG, Pore VS, Mishra NN, Kumar A, Shukla PK, Sharma A, Bhat MK (2009) Synthesis and antifungal activity of 1,2,3-triazole containing fluconazole analogues. Bioorg Med Chem Lett 19:759–763. https://doi.org/10.1016/j.bmcl.2008.12.026
17. Wang XL, Wan K, Zhou CH (2010) Synthesis of novel sulfanilamide-derived 1,2,3-triazoles and their evaluation for antibacterial and antifungal activities. Eur J Med Chem 45:4631–4639. https://doi.org/10.1016/j.ejmech.2010.07.031
18. Buckle DR, Outred DJ, Rockell CJM, Smith H, Spicer BA (1983) Studies on v-triazoles, 7: Antiallergic 9-oxo-1H,9H-

benzopyrano[2,3-d]-v-triazoles. J Med Chem 26:251–254. https://doi.org/10.1021/jm00356a025
19. Giffin MJ, Heaslet H, Brik A, Lin YC, Cauvi G, Wong CH, McRee DE, Elder JH, Stout CD, Torbett BE (2008) A copper(I)-catalyzed 1,2,3-triazole azide-alkyne click compound is a potent inhibitor of a multidrug-resistant HIV-1 protease variant. J Med Chem 51:6263–6270. https://doi.org/10.1021/jm800149m
20. Patpi SR, Pulipati L, Yogeeswari P, Sriram D, Jain N, Sridhar B, Murthy R, Anjana Devi T, Kalivendi SV, Kantevari S (2012) Design, synthesis, and structure-activity correlations of novel dibenzo[b, d]furan, dibenzo[b, d]thiophene, and N-methylcarbazole clubbed 1,2,3-triazoles as potent inhibitors of Mycobacterium tuberculosis. J Med Chem 55:3911–3922. https://doi.org/10.1021/jm300125e
21. De Simone R, Chini MG, Bruno I, Riccio R, Mueller D, Werz O, Bifulco G (2011) Structure-based discovery of inhibitors of microsomal prostaglandin E2 synthase-1, 5-lipoxygenase and 5-lipoxygenase-activating protein: promising hits for the development of new anti-inflammatory agents. J Med Chem 54:1565–1575. https://doi.org/10.1021/jm101238d
22. Ashok D, Srinivas G, Kumar AV, Gandhi DM (2016) Microwave-assisted synthesis and evaluation of indole based benzofuran scaffolds as antimicrobial and antioxidant agents. Russ J Bioorganic Chem 42:560–566. https://doi.org/10.1134/S1068162016050034
23. Ashok D, Srinivas G, Kumar AV, Gandhi DM, Reddy MS (2015) Facile ionic liquid-mediated, microwave assisted green synthesis and antioxidant studies of novel indolin-2-one annulated spirochromanone conjugates. Russ J Gen Chem 85:708–717. https://doi.org/10.1134/S1070363215030305
24. Ashok D, Gandhi DM, Srinivas G, Kumar AV (2014) Microwave-assisted synthesis of novel 1,2,3-triazole derivatives and their antimicrobial activity. Med Chem Res 23:3005–3018. https://doi.org/10.1007/s00044-013-0880-1
25. Ashok D, Kumar AV, Gandhi DM, Srinivas G, Kumari KM, Vijjulatha M, Sridhar B, Prasad E (2015) Synthesis, antimicrobial activity and molecular docking of novel tetracyclic scaffolds incorporating a flavonoid framework with medium sized oxygen heterocycles. Bioorg Med Chem Lett 25:898–903. https://doi.org/10.1016/j.bmcl.2014.12.066
26. Pathoor R, Bahulayan D (2016) Synthesis of large Stokes shift and narrow emission indole-triazole-carboxamide peptidomimetics via MCR-Click strategy. Tetrahedron Lett 57:2360–2366. https://doi.org/10.1016/j.tetlet.2016.04.040
27. Wilkening I, Signore GD, Hackenberger CPR (2011) Synthesis of phosphonamidate peptides by Staudinger reactions of silylated phosphinic acids and esters. Chem Commun 47:349–351. https://doi.org/10.1039/C0CC02472D
28. Hu M, Li J, Yao SQ (2008) In situ "click" assembly of small molecule matrix metalloprotease inhibitors containing zinc-chelating groups. Org Lett 10:5529–5531. https://doi.org/10.1021/ol802286g
29. OSIRIS Property Explorer software, version 2, Actelion Pharmaceuticals Ltd., www.actelion.com; http://www.organic-chemistry.org/prog/peo/
30. Lipinski CA, Lombardo F, Dominy BW, Feeney PJ (2001) Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. Adv Drug Deliv Rev 46:3–26. https://doi.org/10.1016/S0169-409X(00)00129-0
31. Fadda AA, Afsah el-SM, Awad RS (2013) Synthesis and antimicrobial activity of some new benzo and naphthonitrile derivatives. Eur J Med Chem 60:421–430. https://doi.org/10.1016/j.ejmech.2012.11.017
32. Palomino JC, Martin A, Camacho M, Guerra H, Swings J, Portaels F (2002) Resazurin microtiter assay plate: simple and inexpensive method for detection of drug resistance in Mycobacterium tuberculosis. Antimicrob Agents Chemother 46:2720–2722. https://doi.org/10.1128/AAC.46.8.2720-2722.2002
33. Burits M, Bucar F (2000) Antioxidant activity of Nigella sativa essential oil. Phytother Res 14:323–328. https://doi.org/10.1002/1099-1573(200008)4:5<323::aid-ptr621>3.0.co;2-q
34. Cuendet M, Hostettmann K, Potterat O, Dyatmiko W (1997) Iridoid glucosides with free radical scavenging properties from Fagraea blumei. Helv Chim Acta 80:1144–1152. https://doi.org/10.1002/hlca.19970800411
35. Chung KT, Thomasson WR, Wu-Yuan CD (1990) Growth inhibition of selected food-borne bacteria, particularly Listeria monocytogenes, by plant extracts. J Appl Microbiol 69:498–503. https://doi.org/10.1111/j.1365-2672.1990.tb01541.x
36. Azoro C (2002) Antibacterial activity of crude extract of Azadiracta indica on Salmonella typhi. World J Biotechnol 3:347–357
37. Janovská D, Kubíková K, Kokoška L (2003) Screening for antimicrobial activity of some medicinal plants species of traditional chinese medicine. Czech J Food Sci 21:107–110
38. Bishnu J, Sunil L, Anuja S (2009) Antibacterial property of different medicinal plants: ocimum sanctum, cinnamomum zeylanicum, xanthoxylum armatum and origanum majorana. J Sci Eng Technol 5:143–150